

Article

Analysing Cyber Attacks and Cyber Security Vulnerabilities in the University Sector

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Abstract: Universities hold and process vast amounts of financial, user, and research data, which makes them prime targets for cybercriminals. In addition to the usual external threat actors, universities face a unique insider threat from students, who—alongside staff—may lack adequate cyber security training despite having access to various sensitive systems. This paper provides a focused assessment of the current cyber security threats facing UK universities, based on a comprehensive review of available information. A chronological timeline of notable cyber attacks against universities is produced, with incidents classified according to the CIA triad (Confidentiality, Integrity, Availability) and incident type. Several issues have been identified. Limited disclosure of attack details is a major concern, as full information is often withheld for security reasons, hindering institutions' abilities to assess vulnerabilities thoroughly and respond effectively. Additionally, universities increasingly rely on third-party service providers for critical services, meaning that an attack on these external providers can directly impact university operations and data security. While SQL injection attacks, previously a significant issue, appear to have declined in frequency—perhaps reflecting improvements in defences—other threats continue to persist. Universities report lower levels of concern regarding DDoS attacks, potentially due to enhanced resilience and mitigation strategies; however, ransomware and phishing attacks remain prevalent. Insider threats, especially from students with varied IT skills, exacerbate these risks, as insiders may unknowingly or maliciously facilitate cyber attacks, posing ongoing challenges for university IT teams. This study recommends that universities leverage these insights, along with other available data, to refine their cyber security strategies. Developing targeted policies, strengthening training, and implementing international standards will allow universities to enhance their security posture and mitigate the complex and evolving threats they face.

Keywords: cyber crime; UK universities; cyber attack timeline; insider threat; student insider threat; human factors; cyber security awareness; cyber security in higher education



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1. Introduction

In addition to critical research data, UK universities manage personal information on over 2.2 million staff and students annually, along with data on millions of alumni [1]. University networks and computer systems serve tens of thousands of users daily, who access these systems from a wide array of personal devices. The cyber security protections

on these devices vary significantly, and users log in from diverse locations and for varied purposes. Moreover, university websites are intentionally designed to be open and accessible, often disclosing information such as course tutors, support staff, and research personnel. This extensive and diverse surface area exposes universities to a broad spectrum of potential vulnerabilities.

The current technical landscape provides threat actors with enhanced opportunities to conduct cyber attacks, particularly against individuals in roles with access to strategic or confidential data. As a result, universities have become prominent targets for cyber crime, facing threats from diverse adversaries, including state-sponsored actors, cyber criminals, and insider threats [2]. At the time of writing, spear-phishing and social engineering remain the predominant external attack vectors against universities, with ransomware now emerging as the single greatest cause of operational disruption [3]. The types of data sought in these attacks include emails, bulk personal information on staff and students, technical resources, and sensitive research and intellectual property. Although cyber crime represents the most immediately disruptive force impacting universities, state-sponsored espionage likely poses a more significant long-term threat. This form of threat can undermine the value of critical research and erode the UK's strategic knowledge advantage [4].

Aside from the three reports by [2,3,5], there is a notable lack of research on cyber attacks targeting UK universities and the sector's overall cyber security posture. The current study addresses this gap by leveraging publicly available data to outline the types and prevalence of cyber attacks affecting the higher education sector. A discussion of the three foundational reports is provided in Section 2.

In the 2022–2023 academic year, the English higher education sector (excluding institutions in Scotland, Wales, and Northern Ireland) reported a total income of £40.8 billion, representing 1.6% of UK GDP [6]. In 2021–2022, there were 2,182,560 students enrolled in UK universities and a staff population of 233,930 [1]. Fifty-five universities reported enrolments exceeding 20,000 students, while twenty-two had more than 30,000.

With an average enrolment of over 13,000 students per university, disruptions to systems and services can significantly impact student experience and satisfaction, as well as broader societal and economic outcomes. Interruptions to exams and result publication, for example, may severely hinder students' career progression.

Furthermore, service disruptions can adversely affect research grant applications and completion, placing strategically significant research and intellectual property, valued in billions, at risk for the UK research community. Data compromise in these areas has the potential to weaken the UK's academic and economic standing.

It is, therefore, crucial that universities implement robust data protection and cyber security measures. However, effective protective measures require contextual understanding and data-driven insights.

This study analyses 37 recorded cyber attacks against UK universities, with a specific focus on the types of attacks and their associated impacts. The data are presented in the form of a novel chronological timeline, providing a detailed sequence of key cyber incidents. This timeline contributes to existing literature and serves as a basis for future research and facilitates alignment with ongoing studies, setting a foundation upon which institutions can develop informed cyber defence strategies.

The structure of this paper is as follows: Section 2 reviews relevant literature on cyber attacks and cyber crime, including human factors within academic settings, common threats faced by universities globally, and typical cyber defence weaknesses in higher education. Section 3 details the methodology employed to construct the cyber attack timeline. Section 4 presents and analyses the findings, followed by conclusions and recommendations for future research in Section 5.

The significance of this paper lies in its enhanced insights into the cyber threats facing UK Higher Education Institutions (HEIs). These insights are essential for developing adequate protection measures, helping universities safeguard their reputations, maintaining student satisfaction, ensuring curriculum delivery, and potentially reducing insurance costs. Although 55% of HEIs hold some form of cyber insurance, 71% of insured institutions have not made any claims. Many institutions report challenges in obtaining insurance due to a lack of Cyber Essentials certification [3].

2. Background

This section provides an overview of the UK university sector, examining the impact of cyber attacks with a particular emphasis on associated costs and common issues. It concludes with an analysis of threat actors targeting the sector.

2.1. The Landscape

The UK's "education sector" includes primary and secondary schools, further education institutions (such as colleges and sixth forms), and universities and other HEIs offering degree-level courses. While this structure is largely consistent worldwide, terminology can vary, for example, in the United States, the term "college" is often used in reference to universities.

UK universities are attractive targets for cyber attacks for several reasons. They possess substantial and varied income streams and host extensive networks with thousands of service users accessing systems daily from diverse devices. Since the COVID-19 pandemic, universities have increasingly relied on digital systems, moving many services online and broadening their attack surface, which has since been largely retained [7,8].

University infrastructures are fundamentally designed to support open access, promoting widespread participation. However, maintaining open and accessible systems while ensuring security presents a significant challenge, requiring careful balance [9]. Additionally, universities interact with a wide range of research and business partners, meaning that an attack originating within one institution's network could quickly propagate to connected partners. Recognising this risk, Universities UK (UUK) has advocated for a "Defend as One" strategy, emphasising that a strengthened cyber posture at one institution inherently benefits others in the network, including partner universities [9].

Compounding the sector's cyber vulnerabilities is the highly variable technical proficiency of university system users, which ranges from highly skilled to those with minimal IT literacy. As a result, threat actors are increasingly targeting universities, and the number of cyber attacks on institutions has risen sharply in recent years [2,3].

Despite an urgent need for targeted research on the cyber threats facing universities, obtaining sector-specific data has proven challenging. In some cases, data from universities are aggregated with other educational institutions, making it difficult to draw precise conclusions specific to HEIs.

Examples of data conflation issues in a few recent publications include the following:

- JISC CSIRT (Cyber Security Incident Response Team) reports aggregate data on incidents reported by UK universities with data from primary, secondary, and further education institutions, providing a combined overview that includes 2067 compromises, 448 malware incidents, and 650 denial-of-service (DoS) attacks [3].
- The National Cyber Security Centre's breaches survey of educational institutions encompasses 241 primary schools, 217 secondary schools, 44 further education colleges, and 52 HEIs. However, the survey presents findings for the "education sector" as a whole, rather than isolating data pertinent to universities [2].

- JISC's annual cyber impact report on cyber incidents covers both further education institutions and universities, complicating efforts to generalise findings specifically to HEIs [3].
- IBM estimates the average cost of a data breach in the education industry at \$3.65 million for the year preceding their 2023 report. However, this figure combines data for universities, colleges, and 'other training institutes', without isolating the specific costs for HEIs [10].

The present study addresses these gaps by concentrating specifically on cyber attacks affecting UK universities.

UK universities have recently heightened their focus on the complexities of data ownership. While vice-chancellors generally bear ultimate responsibility for institutional data and systems, data responsibility can be less clear in the context of research grants. For instance, "*principal investigators and deans of schools may primarily be responsible for controlling data*" [9]. This nuanced approach to data responsibility underscores the need for robust security measures tailored to the specific challenges faced by universities.

2.2. The Financial Impact of Cyber Attacks on Universities

Cyber attacks on universities can create a range of financial impacts, from delays in publishing results and disruptions to resource access to opportunity costs, decreased student satisfaction, and diminished capabilities in teaching and research. Such attacks can affect not only a single institution but also its partners and, in some cases, national bodies. For instance, if attackers exfiltrate critical research data, the researchers may lose publication opportunities, while the university may suffer reputational damage that could deter investors and impact funding allocations [11]. When the research data pertains to sensitive domains, such as national defence or public health, the implications can escalate to national security risks.

Estimating the financial cost of a cyber attack is challenging in most sectors and is even more complex for universities due to their unique data structures and collaborative networks. Key cost drivers in university cyber incidents include incident response efforts, system restoration costs, legal and regulatory expenses (notably GDPR compliance fines), as well as numerous indirect costs, which this paper does not cover, that are difficult to quantify but nonetheless significant. For example, the loss of educational time could negatively impact the future workforce in high-demand fields, such as medicine and engineering, if essential knowledge is missed due to compromised teaching resources. Moreover, universities may face additional long-term costs in the form of heightened cyber security measures and increased insurance premiums.

Real-world examples illustrate the broad scope of these costs. After a malware attack at the University of California, Berkeley, the institution provided credit monitoring and identity theft insurance to affected students and staff, incurring substantial ancillary costs [12]. Similarly, the University of Giessen was forced to distribute new email credentials manually to 38,000 users after a prolonged network disruption, diverting staff time and resources away from their regular duties [13].

Despite these high costs, there are limited data on the specific financial impact of cyber attacks on UK universities. The most widely referenced figure for the education sector as a whole comes from IBM's report, estimating that in 2022–2023 the average cost of a data breach in "the education industry", a category encompassing universities, colleges, and training institutes, was \$3.65 million [10]. This global estimate, however, lacks UK-specific data and does not disaggregate costs by institution type.

More detailed costs for specific UK incidents were reported by JISC [3], providing a snapshot of incident-related expenses:

- A phishing attack at a UK university in 2019 incurred approximately £65,000 in staff response time and “significant legal costs”;
- A 2020 phishing campaign targeting a college led to around £30,000 in recovery expenses;
- In 2020, a password compromise at a UK university resulted in £40,000 in response costs;
- Another 2020 password compromise led to a compromise of users in other universities leaving around 1000 accounts compromised and around 80 days of response effort.

These examples, whilst now several years old, underscore both the significant costs and the need for further research to inform university administrators and policymakers about the financial risks posed by cyber threats.

Cyber Essentials (CE) Certification

CE is a certification designed to help organisations guard against common cyber threats and signal a commitment to cyber security best practices. Despite the scheme’s potential benefits, many UK institutions are unaware of its provisions, contributing to generally low cyber security practices. In 2017, UK universities with CE certification rated their cyber posture higher on average (6.5 out of 10) than those without (5.8) [14]. The 2020 JISC Posture Survey echoed these findings, reporting average cyber posture scores of 6.4 and 7 out of 10 among UK universities and colleges, respectively, with only 10% and 36% of institutions rating themselves above 8 [3].

A notable example of CE’s importance is the case of the British Library, which held Cyber Essentials Plus certification from 2019 to 2022. However, due to new CE requirements in 2022, the library’s certification lapsed, reflecting the challenge of maintaining compliance when core systems require costly upgrades [15].

Despite its benefits, CE uptake among UK universities is limited. According to the JISC Posture Survey, 34% of universities have no cyber certification, while among those with CE or CE Plus certification, 84% and 68%, respectively, plan to re-certify. The survey notes a recent decline in institutions obtaining CE certification, though more data are required to clarify this trend. For instance, it is unclear whether the reduction in certifications is due to a decrease in initial uptake or challenges in maintaining compliance among previously certified institutions [5].

The following sections will further explore the value of cyber security training and reporting for universities, identify common attack types faced globally, and consider the specific internal risks posed by students and other university-affiliated individuals.

2.3. Example Problems

Numerous studies have exposed areas within university technical infrastructures that require improvement. For instance, a study by [16] examined 500 universities and 5000 university vendors, finding that universities typically hosted an average of 13 unmaintained domains (affecting 5% of surveyed universities), while approximately 10% of the 1500 universities had open Remote Desktop Protocol (RDP) ports, and 48% supported software with known vulnerabilities.

The 2020 JISC Posture Survey [5] revealed that only 16% of universities maintain a fully accurate asset register. This is partly because research project assets are often costed and maintained outside standard infrastructural support, and the Bring Your Own Device (BYOD) policy is widely practised within universities, creating challenges for comprehensive asset tracking.

Further, there has been a decline in universities segregating Wi-Fi networks for students/staff versus visitors, dropping from 92% in 2022 to 77% in 2023. The number of universities employing non-cloud backup methods also decreased from 89% in 2022 to 73% in 2023 [2].

Data breaches represent a significant risk for universities, given that they store data on tens of thousands of current students and staff, along with records on alumni and research data that can include personally identifiable information on research subjects. Crossword Cybersecurity [17] reported that 2.2 million breached credentials belonging to staff or students were found across 100 UK universities. A Redscan report in 2020 indicated that 50% of 134 surveyed universities reported at least one data breach to the Information Commissioner's Office (ICO) in the previous year, while 45% reported account compromises, compared to just 8% in large businesses [2].

Additional research by Chapman [18] highlighted that approximately 86% of university staff were targeted by phishing emails, and 75% reported instances of impersonation of university organisations online. Moreover, 50% of universities experience weekly cyber attacks, and of those, 61% reported adverse outcomes such as data loss or financial losses [2]. Around 63% of universities reported successful intrusions into university-associated devices by threat actors [18].

2.4. Threat Actors

Universities face a wide range of threat actors, including state-sponsored agents, ideological groups, cyber criminals, and hacktivists, all of whom are similarly seen in the corporate sector. Internal threats are specifically addressed in Section 2.4.

According to [9], motivations for cyber attacks on universities generally fall into four categories: extorting ransom payments, stealing research or other valuable data, monetising assets, and disrupting or destroying university infrastructure.

Globally, universities have been targeted by *state-sponsored* actors on numerous occasions. For example, the Chinese government has allegedly targeted the Australian National University [19], North Korea has reportedly targeted Korea University [20], and Iran was implicated in a wiper attack against Bar-Ilan University [21,22]. In 2022, the education sector was the target in 14% of nation-state attacks, with North Korea responsible for nearly a quarter of these incidents [23].

Ideologically motivated attacks against universities are also on the rise. An example is Tsinghua University in China, which saw its website defaced with a photo and audio supporting "holy war" [24], and the University of Botswana, whose logo was replaced by Anonymous's emblem [25].

Cyber criminals, however, likely pose the most immediate threat, particularly due to their use of ransomware to demand payouts. Cyber criminals have targeted university websites in Australia and the United States, in some cases embedding links to "essay mill" sites that promote contract cheating services, or fake contests to harvest and resell student work [26,27].

The Insider Threat

Insider threats in universities stem primarily from staff and students, both of whom can compromise security in unique ways. Research has focused less on insider threats within academia compared to corporate environments, and although no publicly documented cases of staff-perpetrated cyber attacks were found, universities generally keep HR and legal cases confidential. By contrast, multiple instances of student-initiated cyber attacks are publicly documented.

The student threat is unique to the cyber security domain. Students have less—sometimes no—corporate experience but have access to an array of university systems through an array of endpoints that have variable cyber security protection.

The student threat is particularly unique within the cyber security landscape. While students often have little corporate experience, they typically have wide access to various

university systems across multiple devices, many of which have varying levels of cyber protection. JISC [3] highlights “disgruntled students” as a threat, but other motivations, such as curiosity, grade manipulation, personal gain, pranking, skill testing, and revenge, are also common.

Some examples of the student threat are provided in Table 1. In addition to the data contained in the table, there are many examples of student-perpetrated cyber attacks in the public domain including attacks at Austin’s Business School [28] and instances within Kenyan universities where the outstanding fee balances for some students had been altered [29]. Moreover, at the University of Alberta, a student compromised over 3300 accounts [30].

Table 1. Student Cyber Attacks on Universities Globally.

ID	Date	Target	Description	Impact	Ref.
1	14/6/13	University of Purdue	Three students hacked into Purdue professors’ university accounts and changed at least three dozen of their own grades	Personal grades altered	[31]
2	12/1/15	University of Queensland	Student hacked the University grade system and altered his grades	Charged with 14 offences which included forgery, impersonation, and using a restricted computer without consent	[32]
3	23/4/15	University of Birmingham	Student used keylogging software to steal passwords, log onto examination systems, and up his grades, for example, one from 57 to 73%.	Charged under the computer misuse act, jailed for four months	[33]
4	15/1/16	University of Georgia	Student at Georgia tech hacks rival university website and alters content	Charged with computer trespass, asked to apologise and perform community service	[34]
5	11/1/17	University of Iowa	Student changed his grades more than 90 times in 21 months having gained access to the system through a keylogger which stole the user id and password of staff.	Charged with federal computer-hacking charges	[35]
6	11/8/17	Singapore Management University	Student hacked into University computers, altered his final examination grade from D+ to B, final overall grade from B to A- and downgraded the grades of other students.	Charged with the Computer Misuse and Cybersecurity Act and sentenced to 16 weeks jail	[36]
7	11/17	University of South Wales	Student used USB devices to install key logging software on university computers, which he used to obtain login details of 35 university staff members over an 18 month period. Five systems were compromised, including the system used to store exam papers, coursework, and marking. Stolen papers were sold to students.	Student receives a 20 month jail sentence	[37,38]
8	3/8/19	Tufts University, USA	Student allegedly hacked the university grade system and altered her own grades.	Student expelled. Personal grades altered	[39]
9	2019	Maastricht University	Ransomware attack launched on University	University believed to have paid a ransom of around £230,000. Recovery effort involved multiple departments and teams.	[3]
10	3/10/20	Nanyang Technical University	Student hacked into stored value card for hostel pre-paid SIMS, air conditioning, food, drinks and cigarettes.	Stole \$34,000 worth of items, and causing \$80,000 worth of damage. Tee was charged with the Computer Misuse and Cybersecurity Act and sentenced to 10 months jail	[40]
11	5/3/21	University of the Highlands and Islands (UHI)	Cyber criminals from eastern Europe or the Baltic region launched a ransomware attack using various malware components including Cobalt Strike and Ryuk.	Student and staff services disrupted, including the remote access platform to network drives, files and applications, and printing services; facilities closed for a day; systems such as administration and accounting still impacted several months on; system re-build required.	[41,42]
12	9/10/21	University of South Wales	Student hacks university system and makes £20 k selling exam papers, housemate finds students willing to buy the exam papers.	First student makes £20 k in sales, jailed for 20 months, second student receives suspended 9 month sentence.	[43]

Further data on school-level cyber attacks are provided in Table 2. Though outside the focus of this paper, it is worth noting that the primary motivation among school students for cyber attacks is grade manipulation.

Table 2. Student Cyber Attacks on Schools.

ID	Date	Target	Description	Impact	Ref.
13	19/6/08	Tesoro High School, USA	2 high school students hack school computer to change their own grades	Charged with 69 felonies including 34 counts of altering their own grades and those of others	[44]
14	5/5/14	Miami-Dade County Public School	Student hacked into grade system and changed his own and 4 other student grades	Personal grades altered	[45]
15	5/1/17	University of Alberta	Student deploys malware and compromises 304 university computers and 3323 staff and student passwords.	Account access compromised	[30]
16	3/12/17	Tenafly High school (New Jersey)	High school student hacks into school student system, changes grades, and submits an Ivy league application	Personal grades altered	[46]
17	14/5/18	Ygnacio Valley High School in Concord	Student altered his own grades, and reducing the grades of others.	Charged with 14 felonies	[47]
18	9/8/19	Blackboard VLE	US high school student, hacked Blackboard because ‘he was bored’. Student discovers test grades, medical records, café balances, cryptographically hashed passwords, and photos.	Student reports the bugs to Blackboard who addressed the bugs imminently. Student was suspended for 2 days but no legal action was taken.	[48]
19	17/12/19	High School in Maryland	A high school student hacks into school network. Maintains access for over a year to a high school IP TV network and around 5962 student records.	Eventually the student disclosed himself to the school IT department. No sanctions were applied as the student disclosed himself.	[49,50]
20	08/20	Unnamed UK college	Cyber attack launched on results day	Students cannot access results online, results must be emailed out	[3]
21	3/9/20	Florida school district	16-Year-Old Arrested after eight DDOS attacks on the School’s Online Learning Systems.	Charged with “computer use in an attempt to defraud”	[51]
22	9/9/22	Exam department in Sri Lanka	An unnamed Information and Communication Technology student steals the results of 270,000 A-level students, and enables access to them through his own web portal.	Confidentiality breach	[52]

3. Data Gathering Methodology

This analysis encountered a shortage of academically rigorous data on cyber attacks targeting universities, relying instead on web-based sources and media reports. Although sources like [2,3,5] identify key issues, the data from these reports often lack specific details about attack types and tactics.

A solid foundation of literature exists on topics such as online cheating and the security risks in examinations [53–55]. However, since this study focuses on general cyber attacks within university infrastructures, studies that only pertained to exam security and academic dishonesty were excluded.

Following a similar methodology to [7], data were collected on each cyber attack from multiple reputable sources. Traditional repositories like Web of Science and Scopus were used alongside reputable news sites and industry blogs, including BBC, Reuters, and cyber security-focused platforms. Critical evaluation processes typical in cyber-intelligence operations were employed, weighing conflicting reports and including only the most credible information. This review process was limited to English-language sources, and incidents lacking reliable attribution were excluded from the study.

Cyber security news platforms have proven particularly useful for tracking cyber incidents. Trusted sites for accurate reporting on cyber attacks included: [computerweekly.com](https://www.computerweekly.com), [cybersecurity-insiders.com](https://www.cybersecurity-insiders.com), [cyware.com](https://www.cyware.com), [infosecurity-magazine.com](https://www.infosecurity-magazine.com), [itgovernance.co.uk](https://www.itgovernance.co.uk), [itpro.co.uk](https://www.itpro.co.uk), [recordedfuture.com](https://www.recordedfuture.com), [securityweek.com](https://www.securityweek.com), [thehackernews.com](https://www.thehackernews.com), [theregister.com](https://www.theregister.com), [threatpost.com](https://www.threatpost.com), and [zdnet.com](https://www.zdnet.com). The search was restricted to sources published until September 2024, and the bulk of data gathering took place between June and September 2024. As these sites focus on cyber security, Google’s “site:” function was used with these trusted sites to conduct keyword searches, employing terms like “university”, “universities”, and “Higher Education” to pinpoint relevant content.

The data collection employed a wide range of keyword combinations, such as “university” and “cyber attack”. Table 3 shows all the search terms developed for four keyword themes we had. Table 4 shows the exact number of results for a selection of search terms and six databases we used.

Table 3. Literature search keywords and related terms.

Keyword	Related Term
University	Higher Education, universities, HE, HEI
Cybercrime	Cyber crime, cyber criminal, cyber threat actor, cyber breach
Cyber attack	Malware, ransomware, hack, phishing, DDoS
Cyber security	Cybersecurity, information security'

Table 4. Exact search results per term incl. combination of terms for 6 databases used.

Search Term	WoS	Scopus	IEEE Explorer	Springer Link	Google Scholar	ACM
"university cyber attack"	0	4	0	1	11	0
"university cybercrime"	0	13	0	5	90	1
"university cyber security"	0	78	5	19	282	15
"university cyber crime"	0	0	0	0	4	0
"university cyber criminal"	0	0	0	0	4	0
"university hacked"	0	1	0	8	29	0
"university" & "cyber attack"	0	26,505	1509	12,470	68,500	736
"university" & "cybercrime"	0	12,864	764	7777	83,600	744
"university" & "cyber security"	0	55,958	8309	22,759	304,000	3819
"university" & "cyber crime"	0	5963	234	3330	47,500	295
"university" & "cyber criminal"	0	1485	37	1952	5230	61
"university" & "hacked"	0	4856	222	49,730	103,000	1479
"universities cyber attack"	0	4	0	1	4	0
"universities cybercrime"	0	13	0	5	1	0
"universities cyber security"	0	78	2	19	21	2
"universities cyber crime"	0	0	0	0	2	0
"universities cyber criminal"	0	0	0	0	0	0
"universities hacked"	0	1	0	8	6	0
"universities" & "cyber attack"	0	26,505	23	12,430	8710	76
"universities" & "cybercrime"	0	12,864	7	7508	16,600	129
"universities" & "cyber security"	0	55,958	306	22,414	36,300	444
"universities" & "cyber crime"	0	5963	4	3244	8890	44
"universities" & "cyber criminal"	0	1485	0	1950	767	7
"universities" & "hacked"	0	4856	5	49,647	24,000	162
"Higher Education cyber attack"	0	0	0	0	3	0
"Higher Education cybercrime"	0	0	0	0	3	0
"Higher Education cyber security"	0	2	0	1	27	0
"Higher Education cyber crime"	0	0	0	0	0	0
"Higher Education cyber criminal"	0	0	0	0	0	0
"Higher Education hacked"	0	0	0	0	0	0
"Higher Education" & "cyber attack"	0	572	12	1445	4020	32
"Higher Education" & "cybercrime"	0	793	10	1077	10,600	70
"Higher Education" & "cyber security"	0	2128	111	2562	24,300	220
"Higher Education" & "cyber crime"	0	400	4	508	5110	22
"Higher Education" & "cyber criminal"	0	50	0	246	425	5
"Higher Education" & "hacked"	0	239	3	4367	11,300	81

Inconsistent terminology within the cyber security domain posed a challenge, with different sources using varied terms for similar concepts (e.g., “perpetrator”, “hacker”, “attacker”). This aligns with observations in [7] and highlights the importance of consistent taxonomy, especially in categorising attacker motivations, where varied classifications exist [56,57]. This study’s scope covers a twelve-year period, from 2012 to 2024, providing a dataset for future researchers to build upon and address study gaps.

In cases where lesser-known incidents had limited coverage, data were supplemented with social media sources. Using the platform X (formerly Twitter), posts related to university cyber attacks were searched for using the advanced query: “university cyber attack until: 31 July 2024 since: 1 January 2012-filter:replies”. This allowed the location of relevant information on incidents that may not have appeared in traditional media.

Certain incidents remained partially obscured due to non-disclosure by universities. For example, the University of Hertfordshire attack (Table 5) was suspected to be ransomware-related, but the university did not confirm this publicly [58,59]. Similarly, details surrounding attacks on Glasgow Caledonian University and the University of Sunderland remain undisclosed.

Prior to 2015, very few reported incidents per year have been observed in this research, likely due to underreporting. It is possible that the introduction of GDPR legislation has compelled more institutions to disclose cyber security breaches.

This study also uncovered cases involving university students committing cyber crimes outside academic settings. For instance, a student at Teesside University posed as an Amazon support representative to defraud an elderly woman of £40,000, for which he was sentenced to five months in jail [60]. Another case involved two students allegedly hacking into Donald Trump’s tax records [61]. Although noteworthy, these cases were excluded from this study as they did not involve direct attacks on university systems.

Table 5. A timeline of cyber attacks that have compromised UK universities.

ID	Date	Target	Perpetrator. Attack Type. Impact. Motive	Description	Impact	Ref.
23	2/10/12	Multiple *	H.I.C.F	Team GhostShell—a hacktivist group—runs its’ Project Hellfire campaign. Uses SQL injection attacks and targets weak passwords to steal email addresses, passwords, IDs, and the names of staff and students from universities across the globe in its.	Data is made public in Project WestWind. Personal information and credentials of students and staff stolen and posted online.	[62]
		* University of Cambridge; Imperial College London; University of Edinburgh; Manchester University; University of Nottingham; University of Sheffield.				
24	21/5/15	University of London	Ω.D.A.Φ	DDoS attack launched just before students sat their finals.	Moodle inaccessible for over four hours.	[63,64]
25	7/12/15	Multiple universities (unspecified)	H.D.A.Φ	DDoS attack launched on the Janet network, which impacted the majority of UK universities. Threat actors adjusted their point of attack based on JISC’s Twitter updates informing universities of their progress to stabilise the network. Attack lasted four days.	Eduroam Wi-Fi network disrupted (no Wi-Fi access); access to .ac.uk domains disrupted (websites down).	[65,66]
26	13/6/16 5-day attack lasting up to and including 17th	University of Greenwich	Ω.H.C.Φ	Threat actor claiming to be disgruntled former student gains full control of the database structure and data.	Data for 21,000 people stolen and put on the dark web, including data on full names, contact details, and physical and mental health issues. University website defaced. Stolen files posted on dark web. Significant distress caused to students and staff members. ICO investigation reports serious breach and fines £120,000.	[67,68]
27	Dec./16	Multiple universities *. * University of Cambridge; University of Oxford; University of Chester; University of Leeds; University of Glasgow; University of the Highlands and Islands; University of the West of England; University of Edinburgh.	H.I.C.F	Russian-speaking hacker Rasputin performs SQL injection attack on at least 60 global university and government organisation websites.	Access credentials sold on Darkweb. Databases containing personal information compromised.	[69,70]
28	14/7/17	University College London, Ulster University	H.P.A	A ransomware attack in which an email link leads to a compromised host in turn leading to ransomware from the Mole family delivered through the Astrum exploit kit. The exploit kit is a known tool of the AdGholas group.	The cyber criminals demanded a ransom of 3 BTC in exchange for the decryption key. Network and shared drives. Network and shared drives had to be locked down with read-only access; student management system restricted; infections had to be contained; encrypted drives had to be recovered from data backup.	[71–73]
29	May/18	Multiple universities (unspecified)	H.P,M.C.F	Spear-phishing emails sent as part of the STOLEN PENCIL campaign containing a link that would download a malicious Chrome extension, leading to credentials being stolen largely from biomedical engineers.	University login credentials compromised. IP exfiltrated.	[74,75]
30	Aug./18	Multiple universities (unspecified)	H.P.C.F	Academics and students at multiple unspecified UK universities were targeted by the ‘Silent Librarian’ campaign. Phishing emails, often spoofing a university’s library with stolen branding, would direct people to a login page on a false domain that harvested login credentials. After submitting credentials, victims are redirected to their actual university library page, often leaving no indication that information had just been phished.	University login credentials compromised; valuable research stolen.	[76,77]
31	11/9/18	University of Edinburgh	H.D.A.Φ	University campus network suffers a DDoS attack during Freshers’ Week when hundreds of new undergraduates were starting their courses.	University’s website, wireless connections, and other student services down for over 24 h.	[78,79]

Table 5. Cont.

ID	Date	Target	Perpetrator. Attack Type. Impact. Motive	Description	Impact	Ref.
32	2018 to 2019	unnamed 144 US and UK universities	H.P,M.C.F	Cybercrime group launches the <i>silent librarian</i> attack	Theft of around \$3.4 billion worth of IP, 31.5 TB of data, and compromised 7998 university accounts.	[3]
33	2019	The University of Warwick	H.H.C.I.F	Cyber criminals exploited a remote-viewing software that had been installed by a staff member to gain access to the administrative network. An internal report was unable to identify what data had been stolen due to the state of cyber security protections that had been in place.	University suffers multiple data breaches; Sensitive information on students, staff, and volunteers for research studies stolen; ICO investigation that made 144 recommendations including 23 that were classed as urgent; ICO recommended the removal of the chair of the university's data protection privacy group.	[80–82]
34	19/7/19	Lancaster University	H.P.C.F	The university was exploited by a phishing attack, although further details on the attack are unavailable. Names, addresses, phone numbers, and email addresses were stolen from undergraduate student applicant data records. Student records and ID documents were stolen from the student records system.	Fraudulent invoices demanding large amounts of money sent to some undergraduate applicants; distress to students affected.	[83–85]
35	31/7/19	University of York	Ω.Θ.C.Φ	A “malicious data breach” was reported, although further details on the attack are unavailable.	Perpetrator accesses the administrative records of 88 students and views private data. Further data on 4440 students were downloaded.	[86–88]
36	Aug./19	Multiple unnamed universities	H.P.M.C.F	Second wave of the <i>Silent Librarian</i> campaign targeted multiple unspecified UK universities once more, with minor variations to the social engineering techniques and infrastructure used in the previous year.	University login credentials compromised; valuable research stolen.	[89,90]
37	2019	Unnamed UK university	Ω.Θ.C.Φ	Cyber attack	Breach of around 4,100 student records.	[3]
38	May/20	Cloud software provider Blackbaud * Impacted HEIs included Birmingham, De Montfort, Strathclyde, Exeter, York; Oxford Brookes, Loughborough, Leeds, London, Reading, UCL, Oxford	H.R.A.F	Blackbaud provides services for a number of UK universities *, suffered a ransomware attack. They prevented clients being locked out of their own data and servers by paying the ransom. However, the cyber criminals extracted a copy of a subset of data. Some financial data and passwords were stored in cleartext.	Personal data of staff, students, alumni, and supporters stolen; bank account information compromised; user passwords compromised.	[91,92]
39	11/5/20	The University of Edinburgh	H.H.A.F	The ARCHER supercomputer was infected with cryptocurrency mining malware. It was exploited through compromised SSH credentials.	Supercomputer used to mine cryptocurrency for criminals; supercomputer shut down for several days.	[93]
40	28/8/20	Northumbria University	H.R.A.Φ	Cyber criminals infiltrate the university's networks, steal data, and encrypt machines before demanding a ransom payment.	University's clearing service unable to take calls; exams postponed; campus closed for seven days; students unable to access their online student portal or other platforms.	[94,95]
41	Sep./20	Multiple universities * * Glasgow Caledonian University; University of Bristol; University of Cambridge; University of York; University of Kent; King's College London; Queen Mary University of London; University of Lincoln	H.P.M.C.F	Third wave of “Silent Librarian” campaign	University login credentials compromised; valuable research stolen.	[96,97]

Table 5. Cont.

ID	Date	Target	Perpetrator. Attack Type. Impact. Motive	Description	Impact	Ref.
42	04/9/20	Newcastle University	H.R.A.F	DoppelPaymer steals university backup files, encrypts systems, and threatens to release the data unless ransom is paid. Data included a full list of students, their departments, courses, and student numbers, and home addresses, phone numbers, and personal email addresses for students.	Stolen files posted online; many university IT systems not operational for weeks, including relating to timetabling, building access, and smartcard issuing.	[98,99]
43	13/12/20	London Southbank University (LSBU)	Ω.D.A.Φ	Cyber attack disables Moodle, student hub and other university systems.	Students unable to access Moodle, their student hub, recorded lectures, or academic resources for over a month; University-wide institutional login systems down.	[100,101]
44	13/2/21	The Chancellor, Masters and Scholars of the University of Oxford	H.Θ.I.F	Cyber criminals exploit a poorly secured server in the Strubi labs to gain access to lab equipment, pumps and pressure tools.	Biochemical preparation machines compromised; Cyber criminals attempted to sell access to the lab's systems.	[102,103]
45	24/2/21	Queen's University Belfast (QUB)	Ω.Θ.A.Φ	Systems targeted in cyber attack, full extent does not appear to have been revealed, but institution disabled access for staff 'as a precaution'.	Regular outages still occurred in some university systems three months after the incident, causing delays across the university; Health Trusts disconnected a number of services connected to QUB and stopped accepting messages from QUB email addresses, disrupting communication between staff and students working on wards or other trust settings.	[104,105]
46	17/3/21	University of Northampton	H.R.A.F	A ransomware attack at the university caused serious server problems, with phone lines and IT systems taken down.	Six-day outage of IT, telephone, and server networks; students unable to access study materials for on-campus and online induction; access to online teaching disrupted; email systems unavailable.	[106,107]
47	13/4/21	University of Portsmouth	H.R.A.F	Ransomware caused disruption to IT services, with technical disruption continuing over a couple of weeks.	University campus closed for over a week; start of new term delayed; key IT systems taken offline; staff and students unable to log onto network.	[108,109]
48	14/4/21	University of Hertfordshire	H.Θ.A.Φ	The university experienced a cyber attack that impacted several systems including those in the cloud.	Large parts of the network are disconnected; access to cloud-based services, including Canvas, Office 365, MS Teams, and Zoom, blocked; all online classes cancelled; VPN, data storage, and email access all offline.	[58,59]
49	14/5/21	Glasgow Caledonian University	H.Θ.A.Φ	Cyber attack impacts university systems.	Still not clear what happened. However, the university suggested that systems were closed down as a precautionary measure.	[110]
50	12/10/21	University of Sunderland	H.Θ.A.Φ	A cyber attack affects university systems.	University website, IT systems—including the university's records systems—and telephone lines disabled. Online classes cancelled, processing of marks and awards delayed, staff members had difficulty accessing their emails.	[111,112]
51	Feb./22	Imperial College London, UK	Ω.Θ.C.F	Threat actors accessed student names, courses, and owed amounts.	Threat attempted to defraud students into paying debts into a fraudulent account.	[113]

Table 5. Cont.

ID	Date	Target	Perpetrator. Attack Type. Impact. Motive	Description	Impact	Ref.
52	15/3/22	Heriot-Watt University	H.M.A.Φ	The university was breached with a malware attack on its VPN, which led to on-premises infrastructure being taken down for over a week, including access to the VPN for remote login, Oracle R12 finance systems, staff shared areas, and some staff and student directories.	All users of systems had to be moved to alternative systems. Remote learning was hampered for over a week.	[114,115]
53	6/6/23	The University of Manchester	H.H.C,A.F	Threat actors apparently did not encrypt data, just exfiltrated them.	Data were stolen, but systems were kept online, ransomware apparently not involved.	[116,117]
54	1/7/2023	University of the West of Scotland	H.R.C,A.F	University suffers a ransomware attack by Rhysida, leading to the theft of confidential information and a demand for 20 bitcoin.	50% shutdown of systems, laptops, and student submissions. Stolen data included bank details, national insurance numbers, and internal documents.	[118]
55	2/8/2023	BPP University London	Ω.D.C,A.Φ	A cyber attack caused by an unnamed third party which had accessed the University systems causing significant availability issues.	21,000 students unable to access coursework as exams approached.	[119]
56	28–29/10/23	British Library	H.R.C,A.Φ	The British Library was impacted by a ransomware attack which disrupted online services, public WiFi, and the EThOS system. This delayed interlending systems as well as access systems. Parts of the system were deleted or encrypted, and user account data were stolen and published on the dark web.	10,000 s of University students and other users unable to access services.	[15]
57	19/2/24	University of Cambridge	H.Θ.A.Φ	Anonymous Sudan, a hacktivist group claimed responsibility for a DDoS attack on the Janet network which disrupted internet connectivity at the Universities of Cambridge and Manchester. The attackers cited the “UK’s continued support for Israel.” The attack affected systems at both universities, causing delays in email services, problems accessing Moodle and other essential IT services, and impacting internet access.	Full extent unknown but significant accessibility issues reported	[120]
58	19/2/24	University of Wolverhampton	Ω.Θ.A.Φ	Cyber attack renders systems inaccessible, including access control systems enabling physical access to university	Staff and students forced to work at home for 2–3 days	[121]
59	27/6/24	Cambridge University Press and Assessment	H.R.C,A.F	Documents/papers stolen allegedly asking for ransomware as well as threatening to sell on the dark web.	Data theft, possible extortion too	[122]

Perpetrator key: U:University Student; H:Hacker; O:Other; S:High school student; Ω:Unknown; Attack type key: D: Denial of service; E:Extortion; H:Hacking; I:SQL injection; K:Keylogger; M:Malware; P:Phishing (or smishing); R:Ransomware; Θ:unknown; Impact key: C:Confidentiality; I:Integrity; A:Availability; Motive key: F:Financial; O:Other; P:Prank; G:Personal gain; Φ:Unknown; Continued on next page.

4. Results and Analysis of Cyber Attacks and Associated Risks

The data analysis for this study, summarised in Table 5 and Figure 1, documents 37 cyber attacks affecting UK universities. This list, while not exhaustive, offers a representative view of the types and impact of cyber attacks in the sector. The table's row numbers correspond with entries in the figure for ease of reference, depicting attack events alongside the corresponding elements of the CIA triad: Confidentiality (C), Integrity (I), and Availability (A).

The Confidentiality, Integrity, and Availability (CIA) triad is a core security framework: **Confidentiality** ensures data secrecy, **Integrity** safeguards against unauthorised modifications, and **Availability** ensures dependable access to systems and information when required, maintaining trust and functionality.

These classifications were determined based on the nature of the compromise indicated in reports, with the assumption that multiple elements of the CIA triad could be affected in a single attack (e.g., impacting both Integrity and Availability). However, no instances of simultaneous compromise across multiple CIA categories were clearly identified in the dataset.

The analysis categorised attacks by method when specified in the reports. These included the following:

- Phishing (P): Phishing was identified in cases such as the “Stolen Pencil” attacks, where the attackers employed targeted phishing techniques to acquire sensitive information.
- SQL Injection (I): Certain attacks involved SQL injection, compromising databases and exposing sensitive information.
- Denial of Service (D): Attacks causing system unavailability were classified as Denial of Service (DoS), although the specific techniques varied.
- Ransomware (R): Ransomware attacks, which encrypt data and demand payment for decryption, have posed a persistent threat to universities.

In cases where the attack vector was unclear or unspecified, the attack was labelled as a general “hack” (H). For example, the BPP University attack (Row 33) lacked detailed information on the specific methods used and thus received this general classification.

While exact attack dates were not critical to the scope of this study, efforts were made to approximate dates where possible. Dates were gathered primarily from news sources and university announcements. Where precise dates were unavailable, the attacks were marked by month. For attacks on the same day affecting multiple institutions (such as Rows 35 and 36 on 24 February 2019), these are listed separately due to the lack of information confirming a shared threat actor or attack strategy.

This section provides an initial high-level analysis of the data, with more detailed breakdowns of attack types to follow. This study indicates the presence of a wide range of cyber threats to university systems, with different attack methods and various CIA impacts. Each entry in the timeline highlights the diversity of threats faced by universities, including prevalent phishing and ransomware attacks, and underscores the critical need for cyber resilience in the academic sector.

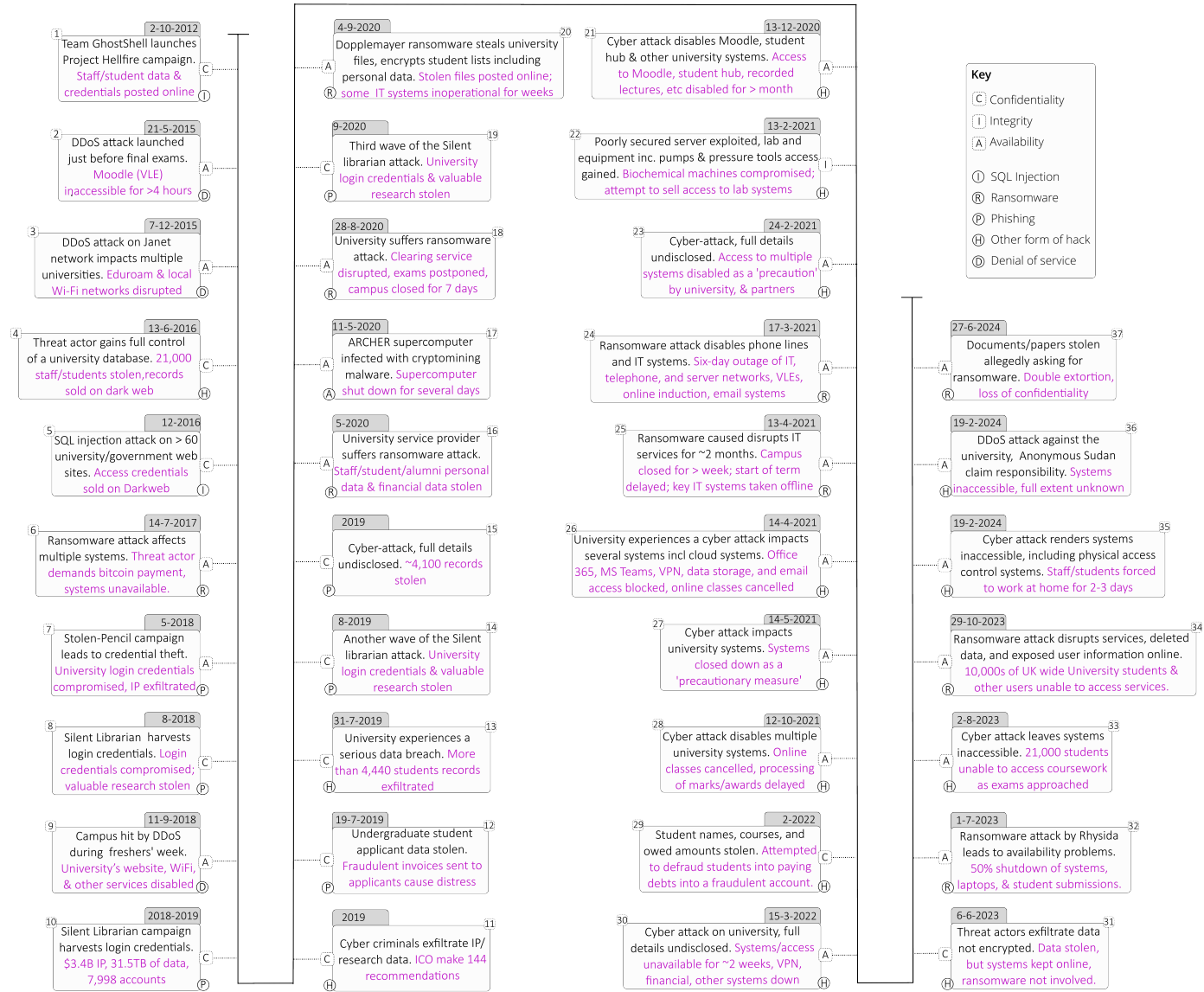


Figure 1. Timeline of key cyber attacks on UK Universities.

4.1. Grey Intelligence

The initial indication of a cyber attack against a university is often observable through student reports on social media, where potentially tens of thousands may comment on disruptions, hinting at the presence of an “issue”. While universities often confirm a cyber attack shortly afterwards, detailed and precise information about these incidents remains scarce. Certain attacks, represented by Rows 2, 13, 15, 21, 23, 27, 28, 29, 33, 35, and 36, lack even basic publicly accessible information. This aligns with findings from JISC, which noted that while its CSIRT received thousands of incident reports, the nature and depth of these incidents are not disclosed and could range from minor issues to complex cyber attacks [3].

The reluctance of universities to release attack specifics is understandable; disclosing details could expose additional vulnerabilities or place partner institutions at further risk. An exception to this trend is the British Library attack (Row 34) [15], which was reported in detail by the institution. This level of transparency provides a rare insight into the anatomy of an attack on an educational institution.

4.2. Third-Party Cyber Attacks on Universities

The significance of third-party risks in the university sector is highlighted by cyber incidents affecting external entities critical to university operations. As noted in [9], managing the cyber security of the university supply chain is essential, given the interconnected nature of institutional networks. A 2020 case documented by JISC [3] described an attempted fraud involving a university supplier, illustrating how attacks targeting suppliers can expose associated institutions: “There is also evidence of institutional attacks leading to the compromise of suppliers and partners via a chain starting at a single institution”. The same report highlights the case of a cyber attack in 2020 wherein cyber criminals attempted to defraud a university supplier. The cyber attack on the British Library (Row 35) and on Queen’s University Belfast (Row 24) exemplify this concern. In the British Library case, service interruptions affected hundreds of educational institutions across the UK and globally. This attack contradicts NCSC’s assertion that universities generally maintain more robust supply chain risk management than private businesses and charities [2].

4.3. SQL Injection Attacks

SQL injection, a once common vector, appeared in the dataset primarily in earlier years (Rows 1 and 5, from 2012 and 2016, respectively) and is absent from later entries. This could be attributed to improvements in web application and database security practices, such as adopting parameterised queries, enforcing strict input validation, and leveraging modern web application frameworks that include built-in protections against injection attacks. These advancements have likely diminished the effectiveness and prevalence of SQL injection as a viable attack method in recent years.

4.4. Phishing

Threat actors frequently exploit societal events to manipulate human behaviour. Notably, the COVID-19 pandemic was linked to a surge in cyber crime reports across multiple sectors [7]. Similarly, phishing campaigns targeting students increase at the start of the academic year, with fraudulent emails offering “free grants” or requesting bank details updates to facilitate loan payments [123]. Although these cases are numerous, this study omits them to remain focused on cyber attacks directly targeting UK universities.

Targeted campaigns like *Silent Librarian* have compromised the credentials of academic staff and students worldwide, including those in the UK, through spear-phishing emails crafted with sophisticated social engineering techniques [96,123]. These campaigns have

led to the theft of university credentials. Likewise, campaigns like *Stolen Pencil* deployed malicious Trojans through spear-phishing aimed at specific fields such as biomedical engineering [75,124,125].

The prevalence of phishing as an attack vector is well-documented. The NCSC [2] characterises phishing as “by far the most common type of breach or attack identified”. JISC [5] ranks phishing and social engineering as the second-highest threat to UK universities, while IBM [10] also reports phishing as the second most frequent attack vector. During a recent period, 144 phishing incidents were reported to JISC CSIRT, placing phishing eighth in overall incidents, likely reflecting a trend where universities report phishing compromises rather than mere attempts [5].

A total of 144 phishing incidents were reported to JISC CSIRT [5] placing phishing eighth in overall incidents. This likely reflects a trend where universities report phishing compromises rather than mere attempts.

Phishing continues to be a dominant method for initial access in cyber attacks, with universities collectively receiving millions of phishing emails annually [126]. The analysis of the dataset identified six cases where phishing was the method of entry (Rows 8, 10, 12, 14, 15, and 19). However, phishing is likely involved in more attacks than this dataset confirms.

Campaigns like *Silent Librarian* (Row 8, 10, and 19) and *Stolen Pencil* (Row 7 and noted in [127] the following year) have reappeared multiple times, suggesting that universities globally remain vulnerable to these threats.

To mitigate phishing risks, many institutions are implementing multi-factor authentication (MFA), with 44% of universities in [3] reporting MFA deployment across relevant systems and 80% applying it to some or all students. While MFA represents a significant step in securing authentication processes, vulnerabilities remain due to continued reliance on single-factor authentication in some systems and the occasional use of email addresses as a second factor, where credentials are often shared across platforms.

4.5. DDoS

Distributed Denial of Service (DDoS) attacks can severely disrupt daily university operations, impacting access to databases, critical digital systems, and physical security control systems. In the 2016–2017 academic year, 41 UK universities experienced DDoS attacks [14]. By 2022, reported DDoS incidents rose to 650 [3]. Though JISC’s 2022 figure combines incidents across both further education and universities, it aligns with the broader trend indicating that DDoS remains a considerable threat.

Among the incidents reported to JISC CSIRT [3], DDoS attacks ranked second in volume, yet they were the lowest priority concern in the [5] University survey, highlighting a possible discrepancy between frequency and perceived threat.

The data show that 26 cyber incidents targeting UK universities impacted availability. Certain attacks appear to have been timed strategically to coincide with busy calendar dates, with some targeting universities during freshers’ week (Rows 9 and 24). These attacks disrupted a variety of critical services, including the following:

- Virtual Learning Environment (VLE) access (Row 2, 21, & 24)
- Website availability (Row 3)
- Wi-Fi access (Row 3 & 9)
- Exam and coursework submission (Row 18 & 33)
- Clearing processes (Row 18)
- Campus closure (Row 18, 20, 25 & 36)
- Class cancellations (Row 26, 28, 28 & 30)
- Delays in result publication (Row 28)

Some universities, like the University of Manchester (Row 32), have mitigated the impact of availability attacks by keeping systems operational during incidents. The University of Manchester's CISO explained that their primary goal was to allow university functions to continue as normally as possible, which contrasts with the more typical approach of taking systems offline during a cyber attack: *"the underlying ambition was to allow the university to function as normally as possible during the incident. This differs from the approach often seen in the education industry, where systems are immediately taken offline"* [116].

4.6. Ransomware

Ransomware has become one of the UK's most significant cyber threats, reaching a level of impact similar to state-sponsored espionage by 2021 [8,128]. Although the 2021 JISC CSIRT ticket data did not explicitly reference ransomware incidents [3], JISC noted a rising trend: 15 ransomware incidents in 2020, 18 in 2021, 19 in 2022, and nine in the first half of 2023 [3]. According to the JISC posture survey, 55% of respondents identified ransomware as the top threat to universities [5].

Recovery from ransomware attacks is often costly. JISC CSIRT reports that some incidents resulted in impact costs exceeding £2 million [3]. NCSC [2] further highlighted the problem, explaining that ransomware attacks have incapacitated entire digital infrastructures, halting online communications, financial transactions, and other critical functions, especially when timed to occur during periods of high activity, such as enrolment, clearing, or examinations.

While law enforcement agencies strongly discourage ransom payments, some universities have paid significant amounts to recover access to systems. For instance, Maastricht University paid nearly €200,000 in bitcoin [129], and the University of California, San Francisco paid over \$1 million to retrieve hacked data [130]. However, payment does not always guarantee full recovery, as demonstrated by Regis University (USA), which suffered months of operational disruption post-payment [131].

The timeline reveals the first publicised ransomware attack on a UK university in 2017 (Row 6), followed by a resurgence of attacks beginning in 2020 (Row 16). Since then, ransomware incidents have become increasingly common, as evidenced by Rows 16, 18, 20, 24, 25, 26, 28, 34, and 36. While ransomware typically impacts system availability, this analysis does not categorise all such cases under availability, as reports often lack definitive details on operational disruption.

5. Conclusions and Future Works

This contribution has thoroughly examined the current landscape of cyber attacks impacting UK universities. Through an analysis of the literature and a detailed timeline of 37 incidents, it offers a foundation for understanding threats specific to the university sector. This research is particularly significant for several reasons:

- **Impacts on University Community.** Cyber attacks disrupt not only the functionality of university systems but also affect staff and students. These disruptions can have operational, financial, and psychological consequences, necessitating stronger defences.
- **The Need to Enhance Incident Response.** The study underscores the importance of efficient incident response strategies to mitigate the disruption caused by cyber incidents. Given the timing and severity of attacks, especially those aimed to disrupt critical academic events, universities must improve response capabilities. Important parameters in these responses include factors such as the number of hours of lecture/teaching delivery cancelled, money lost in failure or delay to fulfil research contracts and/or win potential new bids, and money lost in the loss of potential new students.

- **Training and Cyber Security Culture.** Findings indicate the need for a cyber security culture within academic institutions, which could be achieved through regular training and mandatory security education for both staff and students.

The ramifications of these attacks go beyond operational downtime. Reputational damage can have lasting consequences, particularly when over half of university income relies on student tuition fees, with 65% of prospective students unwilling to apply to universities known for poor cyber security practices [8]. However, despite these risks, only 52% of universities have a formal cyber security strategy [2].

Cyber attacks that become public knowledge frequently expose universities to severe fines and sanctions. For instance, the University of Greenwich incurred a £120,000 penalty from the Information Commissioner's Office (ICO) following a data breach ([132] [Row 4]), and the University of Warwick received 144 ICO recommendations, including a call to remove the chair of its data protection privacy group ([133] [Row 11]). Such incidents underscore the necessity for institutions to invest in robust cyber security practices.

The study also highlights that student and staff well-being is at risk, with incidents leading to stress, health problems, financial loss, and behavioural impacts [134]. For example, the breach at the University of Greenwich exposed 21,000 records ([132] [Row 4]), and the University of Lancaster cyber attack led to fraudulent invoices being sent to applicants ([85] [Row 12]). Such incidents emphasise the need for universities to have robust cyber security measures to protect personal data and alleviate distress for their communities.

Cyber attacks require preparedness and resources for an effective response. While 90% of universities report having dedicated cyber security roles, smaller institutions often face challenges due to limited staffing. With an average of only 3.9 staff members dedicated to cyber security at smaller universities, 29% report struggling to maintain 24/7 incident response coverage [5]. Furthermore, *"many institutions rely on the goodwill of their staff"* for out-of-hours incident management, which JISC suggests should be factored into the financial assessment of cyber security efforts [3].

The 2022 JISC posture survey identified five equally critical dimensions to mitigate cyber security threats: leadership, technical basics, awareness and training, certification, and cyber insurance [3]. This aligns with this study's findings and underscores the need for a comprehensive, multi-faceted approach to managing cyber risks.

The case studies demonstrate that mandatory cyber security training is essential for both staff and students, and it should include regular refresher sessions to maintain awareness. Embedding a culture of cyber security, such as regular communication on security topics, can enhance the effectiveness of these efforts. Given that most students do not receive adequate cyber security training, universities should take a proactive role in preparing students for digital risks, both within academia and as they transition into the workforce. Postgraduates, who often have access to sensitive resources, should be treated with the same level of security and training as staff.

Training should be supplemented with good cyber culture and practices. The study recommends establishing a culture where email communications, particularly from IT and cyber security teams as well as university senior leadership, do not include links or attachments, thus directing users to external resources and reducing the risk of credential theft or malware installation. This approach could mitigate the risk of authority-type spear phishing. However, as outlined by [135] technical measures should augment advice and training designed to recognise phishing.

Existing cyber security policies should be evaluated against international standards such as the ISO 27000 series to identify gaps, especially in disaster recovery and data backup protocols. An overarching information policy that consolidates all data handling

requirements, standards, and best practices would offer a robust governance structure for managing information across the university.

5.1. Limitations

This study has revealed challenges in obtaining comprehensive data on cyber attacks within the university sector. Relying solely on publicly available data, this research provides an initial dataset and a detailed chronology of major cyber incidents. This work offers a foundational view of attack patterns, helping to identify exploitation points and impacts on universities.

Nevertheless, there are limitations to this study. First, the sample size is limited to published incidents, which likely omits numerous unreported or undisclosed cases. The restricted dataset could impact the representativeness of the findings, suggesting that additional data might be necessary to strengthen future research outcomes.

Furthermore, in certain cases, specific details regarding attack characteristics were unavailable. Many cyber attacks encompass multiple phases, such as a ransomware incident that might begin with phishing and include a DoS attack before the deployment of ransomware. However, this dataset does not capture this level of granular detail. Consequently, the nuanced progression and layered nature of certain attacks may be underrepresented in our analysis.

5.2. Recommendations and Further Research

There are several areas in which this research contribution can be further developed.

5.2.1. Better Data

The analysis in Section 2.1 underscores that several influential reports, while informative, often aggregate data across varied educational institutions, making it challenging to isolate trends specific to UK universities. For instance, sources like [3] combine data from schools, further education institutions, and universities, while [2] integrates information from various educational bodies, which could obscure university-specific insights. To equip decision-makers within the university sector with a clearer understanding of cyber security challenges, data specifically focused on universities should be prioritised, with a particular emphasis on sector-specific risk, attack vectors, and impact assessments.

5.2.2. More Detailed Timeline

Further studies could expand upon the current timeline by engaging with universities directly. Conducting surveys and interviews could yield detailed insights into additional potentially unpublished attack data that may not be widely reported in academic or media sources. Universities willing to disclose further details of their cyber incidents could contribute valuable information on attack patterns, timelines, and response measures.

5.2.3. More Research on Costing

Existing literature, such as the [10] report, often merges the financial impact of cyber attacks across a broad “education industry” spectrum, including universities, colleges, and training institutes. This generalised costing fails to provide university-specific financial assessments, limiting the insights available to executive decision-makers. Although some data are available, such as from [3], more contemporary, university-focused research is essential to quantify the real financial implications. Comprehensive cost analysis is critical for prioritising resources and developing a risk-informed approach within institutions. Universities could also consider incorporating regular cost assessments into their own risk management frameworks to better understand potential financial exposure.

5.2.4. Student Threat

Tables 2 and 5 show examples of insider threats posed by students, a topic only briefly explored in Section 2.4. The insider threat posed by students, whether accidental or deliberate, cannot be ignored. Though not covered in depth, the insider threat from students—stemming from motives such as curiosity, personal gain, or retaliation—is a significant risk. No staff-related insider threats were identified in the dataset, but the possibility remains relevant, given the access levels granted to various university personnel. Understanding and mitigating insider threats from both students and staff require further, targeted research.

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